ViT Based Webcam Gaze Tracking for Accessible Computer Interaction

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1. Project Description and Goals

Motivation. People with severe motor impairments can have a hard time using conventional devices like mouse or tracking pads to operate computers and the traditional desktop applications. Computer accessibility is imperative for these people with in order for them to communicate easily and get access to resources and opportunities. Gaze based interaction can offer an easier way for people to interact with the on-screen controls without the precise finger maneuvers. The problem is that current eye-tracking hardware is extremely expensive especially the standard infrared based tobii [5] trackers however the ordinary webcams have poor resolution and introduce noise. My project intends to explore potential applications of training and fine-tuning a vision transformer for supporting practical and gaze-based interaction using a laptop webcam.

Approach. My initial plan is to use an ImageNet pretrained vision transformer (ViT) like DeiT-Small [6] on a dataset like MPIIFaceGaze [7] or Gaze360 to develop a webcam-based gaze tracking and calibration system for disabled people and fine-tune that model. The facial landmarks will be detected using Google's MediaPipe which provides machine learning solutions for live media processing like face detection and pose-estimations [3]. Webcam will capture the face images and these images will be cropped for dimensional uniformity and then normalized in order to account for factors like the camera angle and head orientation. These images will be processed by the MediaPipe framework to extract key features like eye corners and retinal position coordinates.

The fine-tuning process will involve replacing the vision transformer's original image-classification head with the head which processes and predicts the two important gaze angles which are the pitch (vertical axis) and yaw (horizontal axis). I will fine-tune the model on labeled-gaze images to infer the relationship between facial orientation and gaze direction.

The trained model will be integrated into a real-time webcam application with a desktop overlay to help users interact with the computers. Furthermore, a user specific calibration process beforehand will help map gaze angles to the respective screen coordinates. Moreover, I will explore the scope of the exact operating-system input APIs for enabling interactions.

Evaluation. I will evaluate the performance of the fine-tuned ViT model based on three key metrics: Firstly, gaze-estimation accuracy from the MPIIFaceGaze dataset who were excluded from training and measure the mean angular error between the predicted and true gaze directions. I also plan to explore by comparing these results against a ResNet based model [2] to establish baseline and experimental studies that given certain conditions which could overweigh each other.

Secondly, the real-time webcam application will be evaluated based on how accurately it identifies the provided screen region. I will experiment with using a 3 x 3 grid to calculate the percentage of correctly selected regions after the initial calibration phase is completed.

Finally, the pipeline's overall processing speed in frames per second (fps) and latency will be analyzed for responsiveness for the real-time computer interactions.

Goals. My minimum goal is to fine-tune a ViT on MPIIFaceGaze and evaluate its gaze-estimation accuracy on test data and compare the results with a ResNet baseline. I intend to develop this real-time webcam tracker with user tailored calibration and a screen window selection interface to measure accuracy and latency.

Ideally, I would love to extend this into a more comprehensive and functional accessibility tool by implementing trigger events detection, on-screen keyboard usage and operating system input APIs to control various desktop application and customize settings.

Limitations. As much potential gaze tracking system holds for people with motor impairments, the reality is that the there are severe challenges in this particular domain. The prevalent issue being the midas-touch problem as human eyes naturally browse the visual information

in front of them and the system can struggle distinguishing between user actually looking or intending to initiate an action. Moreover, gaze tracking systems face navigation hurdles with actions like scrolling and are incapable of handling tasks which can include multitasking like playing video games where multiple inputs need to be triggered simultaneously.

2. Resources

- **Data:** MPIIFaceGaze dataset containing gaze-estimations along with annotated facial images from 15 participants captured during everyday laptop use. [7]
- **Models:** DeiT-Small, ResNet Architectures
- **Code** I will develop the fine-tuning and evaluation pipeline.
- **Related work:**GazeTR [1], FAZE [4]

3. Group Members and Roles

This will be a solo project by me and I will be responsible for all components of the project: data preparation, model training, evaluation and the live application tracker. Tentative timeline:

- **Oct 16–Nov 8:** data processing, fine-tuning the ViT and establishing ResNet baseline, and dataset evaluation.
- **Nov 9–Nov 25:** calibration setup, experimenting with functionalities and improving the live tracker
- **Nov 26–Dec 8:** Refining existing functionalities and working on additional features. Preparing the final report and demo video

4. Relationship to Background

I have worked on projects involving making accessibility functionalities for disabled people and pose estimations across different hackathons. Also, I was the sole author of an arXiv paper on AI-enhanced virtual biopsies for brain tumor diagnosis in low resource settings where I applied deep learning to train a classification model on MRI images.

5. AI Use Plan

According to the course policy, I intend to use AI (like Claude, ChatGPT) as my assistant and guide to help me learn more. I will write the project code, run the experiments and prepare the final report and video demo by myself. AI can help me debug issues, brainstorm different approaches, help me breakdown literature readings for better grasping concepts and guide me if I am going astray.

References

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